

Consumer Behaviour Analysis

1. Business Problem Statement

A leading retail company wants to better understand its customers' shopping behaviour in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.

Overarching Business Question: *"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimise marketing and product strategies?"*

2. Project Overview & Dataset Summary

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

- **Total Records:** 3,900 transactions
- **Total Features:** 18 attributes covering customer demographics, purchase details, and shopping behaviour.
- **Missing Data Handled:** 37 missing values in the Review Rating column imputed using category medians.

3. Exploratory Data Analysis & Processing

- **Python Processing:** Data preparation, cleaning, and feature engineering were performed using Python (Pandas & NumPy). Columns were standardised to snake_case, new features such as age groups and purchase frequency intervals were created, and redundancy between discount applications and promo codes was resolved.
- **Database Integration:** The cleaned dataset was subsequently loaded into SQL for relational querying.

4. Executive Dashboard OverviewOverall Metrics:



- **Total Revenue:** ₹22.4M
- **Average Purchase Amount:** ₹5,737
- **Number of Customers:** 3.9K
- **Average Review Rating:** 3.75
- **Key Segment Breakdown:**
 - **Subscription Status:** Non-subscribers comprise 73.0% (2,847 customers), while subscribers account for 27.0% (1,053 customers).
 - **Revenue by Category:** Clothing led with ₹10.0M, followed by Accessories (₹7.1M), Footwear (₹4.3M), and Outerwear (₹1.0M).
 - **Revenue by Age Group:** Young Adults led revenue generation at ₹6.2M, followed by Middle-aged (₹5.9M), Adults (₹5.6M), and Seniors (₹5.4M).

5. Structured SQL Insights (Business Transactions)

Analysis Area	Key Findings / Output Summary
1. Revenue by Gender	Male customers generated 157,890 in revenue compared to 75,191 by Female customers.
2. High-Spending Discount Users	Identified 839 customers who utilised discounts while maintaining above-average purchase amounts.
3. Top Products by Rating	Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78) received the highest ratings.
4. Shipping Type Comparison	Express shipping average purchase amount (60.48) slightly exceeded Standard shipping (58.46).
5. Subscribers vs Non-Subscribers	Non-subscribers accounted to 2,847 customers (₹170,436 revenue), while 1,053 subscribers generated ₹62,645.
6. Discount-Dependent Products	Hat (50.0%), Sneakers (49.66%), Coat (49.07%), and Sweater (48.17%) had the highest discount rates.
7. Customer Segmentation	Segmented base into 3,116 Loyal customers, 701 Returning customers, and 83 New customers.
8. Revenue by Age Group	Young Adults led revenue generation (₹62,143), followed by Middle-aged (₹59,197) and Adults (₹55,978).

6. Business Recommendations

- **Boost Subscriptions:** Promote exclusive member benefits and loyalty tiers to convert non-subscribers (currently 73% of the customer base).
- **Customer Loyalty Programmes:** Reward repeat buyers to transition them into the highly stable 'Loyal' customer segment.
- **Review Discount Policy:** Balance promotional discount frequency on high-elasticity items (e.g. Hats, Sneakers) to safeguard profit margins.
- **Product Positioning:** Highlight top-rated products (Gloves, Sandals, Boots) prominently in seasonal marketing campaigns.
- **Targeted Marketing:** Focus acquisition and express-shipping campaigns on high-revenue demographics like Young Adults.